Digitalizing IGF Mobilization at Tema Metropolitan Assembly.

Ghana's **Local Governance Act 2016 (Act 936)** mandates Metropolitan, Municipal, and District Assemblies (MMDAs) to mobilize Internally Generated Funds (IGFs) as the cornerstone of fiscal autonomy and local development. Under Sections 163-172, Assemblies must levy rates, fees, licenses, and other charges through participatory fee-fixing resolutions, maintain proper accounts, and integrate IGF into composite budgets to fund plans, programs, and service delivery.

IGF Line Items under Act 936

Act 936 authorizes these standard revenue heads, which TMA reports in its financials:

Line Item	What It Constitutes
Property Rates	Annual tax on land/buildings based on valuation rolls (size, location, use). Largest potential source when properly assessed.
Basic Rates	Flat/minimum levy on rateable properties for basic services like roads/sanitation. Often supplementary to property rates.
Fines	Penalties for by-law breaches (sanitation violations, unauthorized structures, market offences). Enforced via taskforces/courts.
Fees	Service charges: market tolls/dues, lorry park fees, building/development permits, birth/death registrations, sanitation fees.
Land	Ground rents on stool/family lands, land sale/lease proceeds, royalties from land transactions.
License	Business Operating Permits (BOPs) for shops/SMEs, trader/hawker licenses, liquor/entertainment approvals.
Investment	Bank interest/deposits, dividends, profits from Assembly commercial ventures (e.g., managed markets).
Miscellaneous	Auctions (impounded goods), donations, asset sales, recoveries/reimbursements.
Rent	Leases of TMA properties: offices, market stalls, residential units, lorry parks.

Note on Property Rates: Since January 2023, the Ghana Revenue Authority (GRA) manages property rate collection through the national Myassembly.gov.gh platform (a quick search shows the website to be down) in partnership with MMDAs. TMA provides local valuation data but GRA handles billing and payments. Taxpayers pay via mobile money, USSD codes, bank cards, or online portals. TMA receives 70% of collections (30% covers GRA operations, tech provider, and central costs).

Current Manual IGF Collection Process at TMA

Pre-collection (Planning and Assessment)

The TMA finance/revenue unit compiles revenue items (property rates, market tolls, business operating permits, lorry park fees, etc.) and sets annual IGF targets through the composite budget process. Assessment teams manually value properties and businesses using outdated valuation rolls and paper-based records.

Billing and Demand Notices

Revenue officers print or handwrite demand notices and deliver them physically to taxpayers (households, traders, businesses, transport operators). TMA maintains taxpayer registers in ledger books or Excel sheets, making it difficult to keep accurate, up-to-date records of ratepayers and amounts due.

Field Collection

TMA engages internal revenue collectors and commission-based taskforces to visit markets, lorry parks, shops and communities to collect cash. Collectors issue paper tickets or carbonated receipts from printed receipt booklets; serial numbers are later reconciled at the office.

Lodgement and Recording

Collectors return daily to TMA treasury/finance office to hand in cash and receipt counterfoils for verification. Cash is manually counted, reconciled against receipt numbers, then lodged into TMA's bank account, often with delays that increase leakage risks.

Accounting, Reporting and Enforcement

Revenue data is manually entered into cash books; errors, incomplete entries and under-declaration are common. Weak controls and limited audit trails hinder leakage detection, as reported by Ghana Audit Service. Enforcement relies on physical follow-ups, taskforces and slow court actions.

TMA's Past Digitalization Efforts: Partial Success

TMA deployed an integrated revenue management system around 2017-2018, enabling real-time field monitoring of collectors. By 2019, officials credited it with cutting leakages and boosting IGF (strong 2016-2018 gains, GH¢33.1M or 74% target by Oct 2025).

Evaluation: Operationally successful (better supervision, higher collections), but incomplete, still relies on paper bills, field visits, manual reconciliation, and taskforce enforcement. Property rates are now semi-centralized via GRA's platform. Full end-to-end automation remains absent.

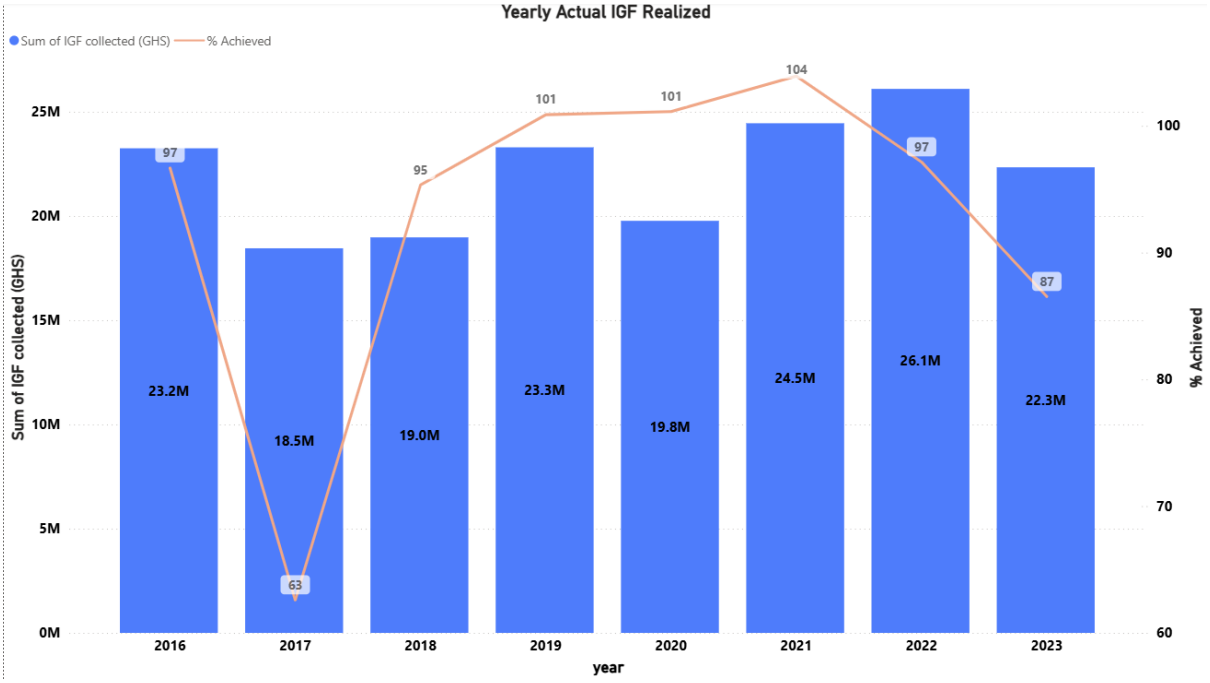
Need for Digitalization

Manual processes cause leakages (under-declaration, cash mishandling), low compliance, and high enforcement costs despite TMA's strong performance. This project automates the entire IGF lifecycle:

- Digital taxpayer database with unique IDs for properties/businesses
- Automated e-billing/SMS notifications replacing paper distribution
- Multi-channel payments (mobile money, USSD, POS, bank) with instant receipts
- Real-time dashboards tracking collections by line item/area/collector
- Defaulter management with automated reminders/escalation before taskforce
- Integration with GRA property rate platform and TMA financial systems

Data Analysis and Findings

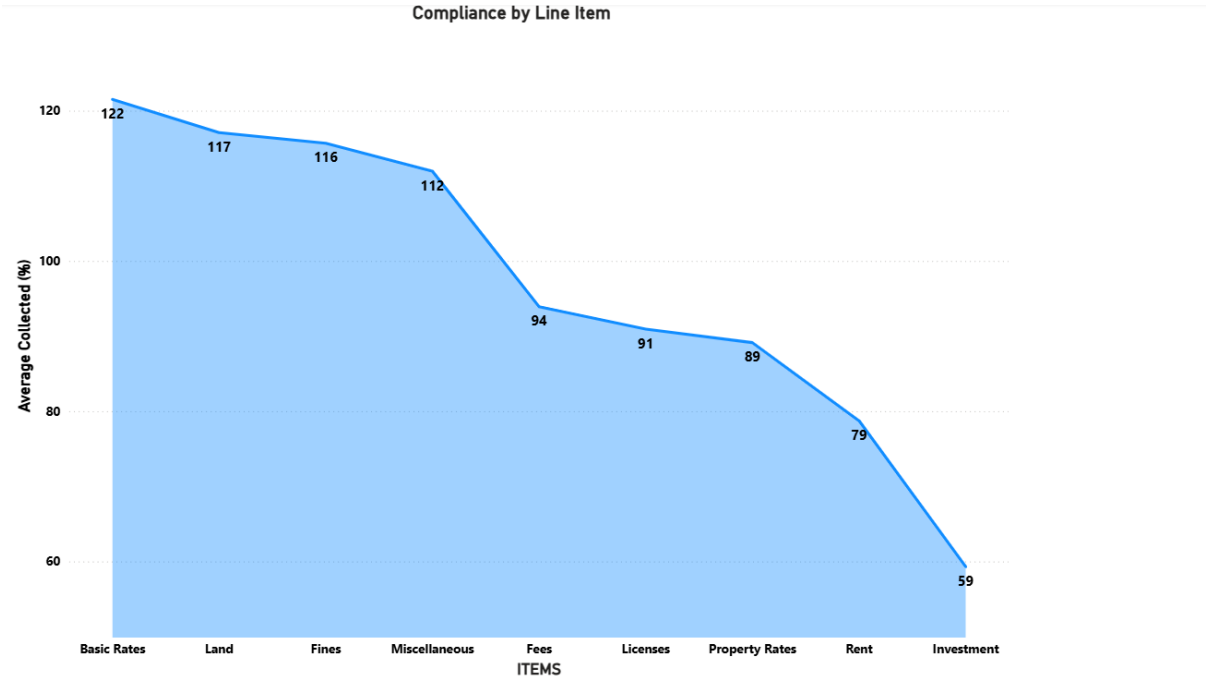
Yearly IGF Actualized



TMA had the highest IGF collection in 2022 with property rates (i.e. collections taxes on lands and buildings) contributing 51% to the revenue pool. It is worth noting that this outstanding performance in 2022 was actually a drop when we put it side-by-side with performance in 2021. This is simply because in 2021 TMA over-achieved its given target by 4% and in 2022 saw a 3% decline with regards to given targets. From the data analysed, property rate is the largest contributor to the revenue and the increased amounts collected can be attributed to an increase in rates and or number of property owners. The dip in 2017 is interesting considering the fact that an integrated revenue collection system where POS devices were used for the recording and billing of revenue was introduced in that year.

This effort of the Assembly to digitalized its revenue collections did not go as planned as we can clearly see a fall in amount collected as well as the percentage achieved when compared with budget given for that year. It can be argued that this initiative started bearing fruits in 2018 as we see a 32% increase in revenue collected when compared with budget given. This rise continues through to 2021 and starts to take a nose-dive from 2022 to 2023. The introduction of the POS device for revenue collection and billing did not solve the problem of revenue collectors taking physical cash from individuals.

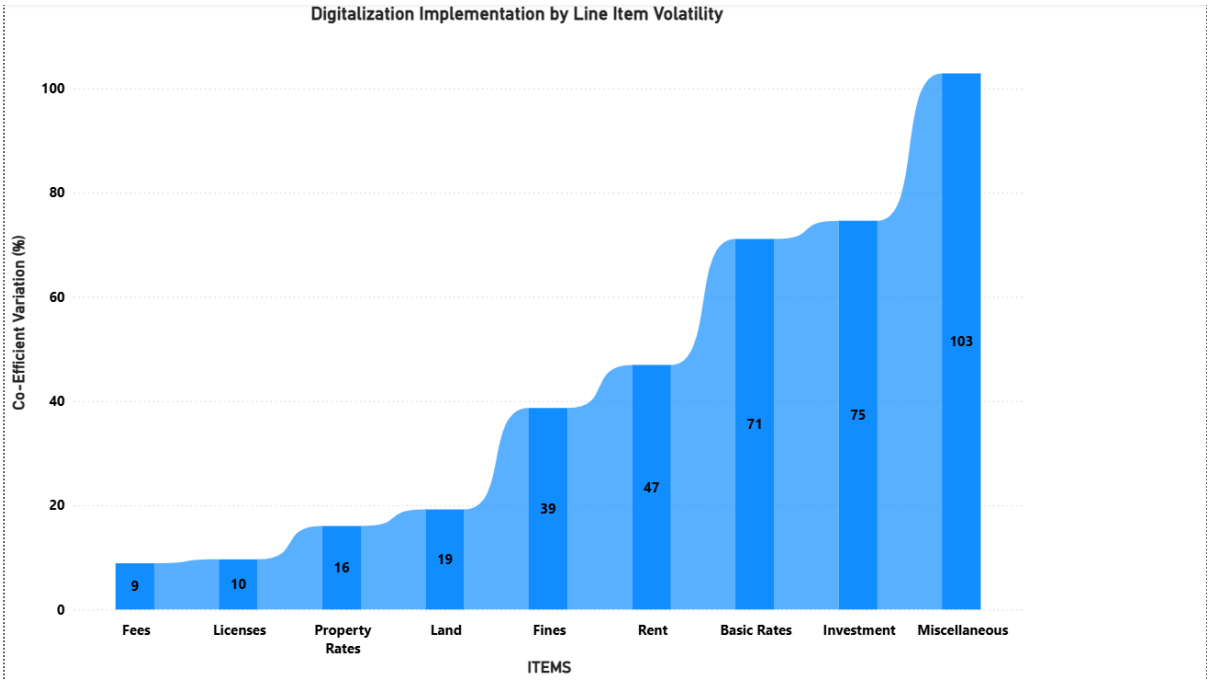
Compliance by Line Item



This graph depicts the percentage of average revenue collections made by TMA on the basis of the line items that make up the Assembly's IGFs. Collections in a way make up for compliance. Items like Basic Rates and Miscellaneous show higher percentages as a result of some abnormal over-achievements in certain years. For example, the budgeted Basic Rates for the entire Assembly in 2020 was GHS 500.00 and an amount of GHS 4,016.50 was actualized (this measures up to an achievement rate of 803.30%). There can be an argument that the Assembly wasn't willing to provide properties for basic services like roads and sanitation in that year because the country was in a lockdown as a result of COVID-19 or this could be a classic example of setting low budgets and over-achieving targets to help boost performance in other areas. The same abnormality occurs in 2017 with Miscellaneous where the revenue actualized was five times the target set. Here, an argument cannot be made that it was as a result of the integrated revenue collection system because the system was mostly used for the collection of property rates.

Investment was not added to the revenue pool until 2021 hence the low average achievement rate. Again, investments do not involve individuals in the Assembly. The remaining line items had average achievement rates ranging from 78% to 115% (Rent and Fines respectively). Surprisingly, these two line-items both over-achieved in 2023 which caused an increase in the average achievement rate. It can be said here that individuals within the Assembly complied well to paying rent for leased TMA properties and stricter fine collections were enforced on those who breached by-laws here the increase in compliance.

Digitalization Implementation by Line-Item Volatility



The digitalization of IGFs work best if it is done in phases. Each phase provides some challenges that would require fine-tuning the system and this is necessary because the architects behind the system learn how best to make the system operate and Assemblies are able to engage in efficient and effective revenue collection and individuals can also hold them accountable for developmental works. This graph measures the rate at which a line-item's collection can change. The goal here is to classify line-items by their volatility and plan system integration phases well with this information in mind. Ideally, it would be best to include line-items with the least volatility in the first phase to see how the system interacts with it and later fine-tune the system to add high volatile line-items.

With this in mind, Fees, Licenses, Property Rates and Land can be include in the phase 1 implementation of the digitalization system as they have rates less than 20% making them highly stable line-items. Fines and Rent are moderately volatile and can be included as time progresses. Basic Rates, Investment and Miscellaneous tend to be highly volatile because the data provided on these line-items are for a 3–4-year period. During these periods they experienced very high over-achievement and have low under-achievements.

Projections

ITEMS	AVG. GROWTH	CARG
Property Rates	4.22%	-4.03%
Basic Rates	22.30%	-
Fees	-3.22%	-4.47%
Fines	86.49%	1.20%
Licenses	5.39%	3.03%
Land	23.25%	6.64%
Rent	43.60%	35.16%
Miscellaneous	29.24%	-100.00%
Investment	24.65%	-

These values represent the average growth rate per line-item and their respective compound average growth rate. Assuming revenue growth will follow this pattern leads to all sorts of mistakes. For example, using the CARG on the largest revenue contributor, i.e. Property rates, suggests a steady decline in revenue mobilized from this source for years ahead. This is highly inconsistent with reality. Hence, to make projections for the upcoming years including when digitalization takes place, we have to classify line-items into fixed and variable revenue contributors in the sense of the elements that aggregate to build up these line-items and the frequency at which their unit rates change. After this we assume the least possible CARG for the line-items and make predictions based on these considering the before and after of digitalization implementation.

Fixed revenue line-items include property rates, basic rates, land, licenses and fees. Semi-fixed revenue line-items may include rent and fines. Variable revenue line-items can include investment and miscellaneous. Average CARG for fixed revenue line-items computes to 0.29%, 18.18% for semi-fixed revenue line-items and -100% for variable revenue line-items. The assumption here is that fixed line-items have low volatility whilst semi-fixed and variable line-items have moderate and high volatility respectively. The negative average CARG of variable line-items is as a result of missing data and abnormal rise and fall in targets and actual values collected. Also, the data shows that revenue for Miscellaneous was not collected for a period of three years consecutively (2020-2023). As a result, these are excluded from the projections.

Going by these assumptions, pre-digitalization IGFs for upcoming years should look like this:

ITEMS	AVG. CAGR	2024	2025	2026	2027	2028	2029	2030

Property Rates	0.29 %	6,807,448.06	6,827,189.66	6,846,988.51	6,866,844.78	6,886,758.63	6,906,730.23	6,926,759.75
Basic Rates	0.29 %	2,406.96	2,413.94	2,420.94	2,427.96	2,435.00	2,442.06	2,449.15
Fees	0.29 %	3,919,269.91	3,930,635.79	3,942,034.63	3,953,466.53	3,964,931.59	3,976,429.89	3,987,961.54
Fines	18.18 %	88,497.32	104,586.13	123,599.89	123,958.33	124,317.81	124,678.33	125,039.90
Licenses	0.29 %	7,807,848.49	7,830,491.25	7,853,199.67	7,875,973.95	7,898,814.28	7,921,720.84	7,944,693.83
Land	0.29 %	3,540,851.68	3,551,120.15	3,561,418.40	3,571,746.51	3,582,104.57	3,592,492.68	3,602,910.91
Rent	18.18 %	279,782.51	330,646.97	390,758.59	391,891.79	393,028.28	394,168.06	395,311.15

For post-digitalization projections, we can make the argument that the average CARG for fixed revenue line-items should increase because the introduction of digitalization which includes automation should increase revenue collection and reduce revenue leakages because there is already an established large base (which is unlikely to experience abnormal increase in elements and rates) and effective and efficient collection and billing methods in place. There should also be an increase in the average CARG for semi-fixed revenue line-items due to the increase revenue collections and decrease revenue leakages the digitalization system promises. To be on a safe side, we assume average CARG for fixed items are now 3% and 20% for variable items.

Revenue projections post-digitalization should look like this:

ITEMS	**AVG. CAGR**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
Property Rates	3.00 %	6,991,396.46	7,201,138.35	7,417,172.50	7,639,687.68	7,868,878.31	8,104,944.66	8,348,092.99
Basic Rates	3.00 %	2,472.00	2,546.16	2,622.54	2,701.22	2,782.26	2,865.73	2,951.70
Fees	3.00 %	4,025,175.00	4,145,930.25	4,270,308.15	4,398,417.40	4,530,369.92	4,666,281.02	4,806,269.45

Fines	20.00 %	89,860.20	107,832.24	129,398.69	133,280.65	137,279.07	141,397.44	145,639.36
Licenses	3.00 %	8,018,829.34	8,259,394.22	8,507,176.04	8,762,391.32	9,025,263.06	9,296,020.96	9,574,901.58
Land	3.00 %	3,636,531.29	3,745,627.23	3,857,996.04	3,973,735.92	4,092,948.00	4,215,736.44	4,342,208.53
Rent	20.00 %	284,091.23	340,909.47	409,091.37	421,364.11	434,005.03	447,025.18	460,435.94

Post-digitalization of TMA's Internally Generated Funds we see an increase in revenue reported by each line-item. TMA experiences 10.22% growth in revenue for the total periods under consideration for projections. This goes to show the benefits of adopting digitalized systems in revenue collection, billing and processing.

Limitations

- There is a great need to recapture individuals in the Assembly to reflect an up-to-date database. The process of recapturing individuals tends to be time and cost consuming.
- Digitalization systems do not mean and end to TMA's revenue collection problems. Issues of corruption may still persist so far as there is the human element.
- Irregularities in TMA's historical IGF data can affect revenue projections digitalization promises.
- Unreliable network connectivity can affect collection and billing processes which can lead to delays.
- There maybe difficulty in keeping up with new businesses with the Assembly.

Recommendations

- In order to ensure high compliance within the Assembly TMA should ensure that developmental projects reflect the collected revenue.
- Individuals should be incentivized in order to register themselves onto the database. Registration could be done through USSD codes and incentives could be reduced rates paid for early or on-time registration.
- Enforcers of revenue payment should not be allowed under any circumstances to collect physical cash from individuals. If the entire process is fully automated these enforcers should be given other tasks to avoid any lay-offs.

- Similar to using USSD codes for individual/business registration, same should be done for revenue collection and billing.

Conclusion

The digitalization of TMA's revenue processes promises revenue growth, accountability and transparency amongst other benefits. For this to reach its full potential, all parties involved in the process have to fully benefit from the system. Phasing implementation would also help to break challenges in bits and solve them head-on. TMA has the potential to be fully self-reliant whilst ensuring that the best of developmental projects and well-thought of services are provided for its individuals and businesses. This is the way forward.